Resultant Forces

Exam code: 0625 & 0972

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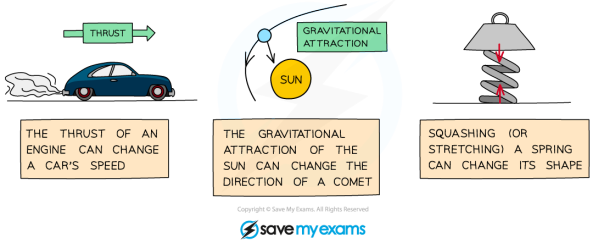
Effects of forces

- A **force** is defined as:

A push or a pull that acts on an object due to the interaction with another object

- Forces can have a variety of effects on an object
- Forces can change an object's
  - **speed**
  - **direction**
  - **shape**
  - **size**

Effects of forces on different objects



A thrust force can cause a car to speed up, a gravitational force can cause a comet to change direction, compression forces can cause a spring to change shape

- The effects of forces on an object often depend on the type of force acting
  - The push force (thrust) of an engine can cause a car to speed up, whilst the force exerted by the brakes (friction) can cause it to slow down
  - The gravitational pull of the Sun on a comet causes the comet to change direction
  - When two opposing forces push on each end of a spring, the spring changes shape (it compresses)

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Resultant forces on a straight line

forces operating on a body
- When multiple forces act on one object, the forces can be combined to produce one net force that describes the **combined action** of all of the forces
- This single resultant force determines:
  - The **direction** in which the object will move as a result of all of the forces
  - The **magnitude** of the net force experienced by the object

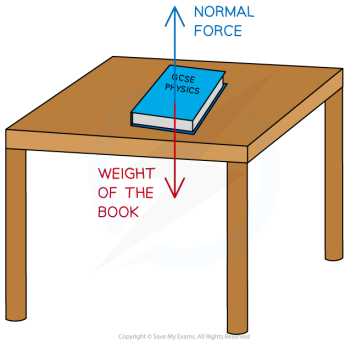
Balanced and unbalanced forces

- The forces acting on an object can be described as **balanced** or **unbalanced**
- Forces are **balanced** if multiple forces act in **opposing directions** with an **equal magnitude** in each direction
  - The effects of the forces then **cancel out**
  - There is **no resultant force** in that plane of direction

- Forces are **unbalanced** if the effects of the forces acting in each plane do **not** cancel out
  - There is a **resultant force** in one or more planes of direction

- A book is at rest on a table
  - The gravitational pull of the Earth on the book (weight) acts in a downward direction
  - The push force of the table on the book (normal contact force) acts in the upward direction
  - The forces are equal in magnitude and opposite in direction
  - The forces are therefore balanced
  - There is no resultant force acting on the book

Zero resultant force on a book resting on a table



A book resting on a table is an example of balanced forces

Calculating resultant force

- Force is a **vector** quantity; it has both **magnitude** and **direction**
- When adding forces together, it is important to assign positive and negative values to show the direction in which the forces are acting
- If a 5 N force acts to the right and a 5 N force acts to the left on an object, then we assign one of the values as

$$\text{resultant force} = 5 + (-5)$$

$$\text{resultant force} = 0\ \text{N}$$

- The forces acting on the object are equal in magnitude and opposite in direction; therefore, they cancel one another out
- This is like two people pushing a box with equal force from opposite sides; the box doesn't move
- If two people push the box from the same side in the same direction, one with a 3 N force and one with a 7 N force, then the forces will add together, and the box will move in the direction of the resultant force

$$\text{resultant force} = 3 + 7$$

$$\text{resultant force} = 10\ \text{N}$$

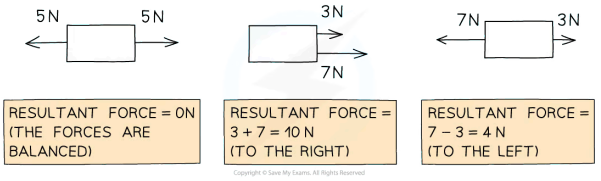
- If two people push the box in opposite directions, one with a 7 N force to the left (negative) and one with a 3 N force to the right (positive), then the forces will add together and the box will move in the direction of the resultant force

$$\text{resultant force} = (-7) + 3$$

$$\text{resultant force} = -4\ \text{N}$$

- In this case, the box will move to the left with a force of 4 N

Zero and non-zero resultant forces



Zero and non-zero resultant forces acting on three objects

Worked Example

Calculate the magnitude and direction of the resultant force in the diagram below.

Answer:

Step 1: Assign a direction to the forces

- Forces acting to the right are positive
- Forces acting to the left are negative

$$\text{resultant force} = (14) + 12$$

$$\text{resultant force} = -2\ \text{N}$$

Step 3: State the magnitude and direction of the resultant force

- The resultant force is 2 N to the left

Examiner Tips and Tricks

Mathematically, it doesn't matter which direction you assign to be positive or negative, as long as you are consistent throughout your calculation.

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